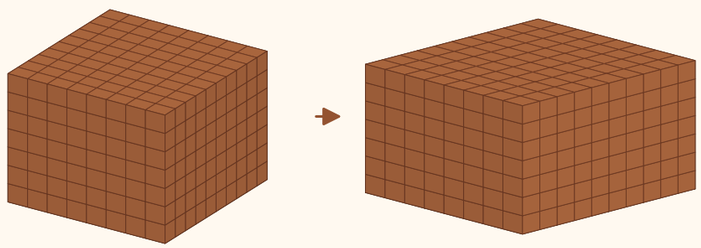
A local solution of Ericksen's problem

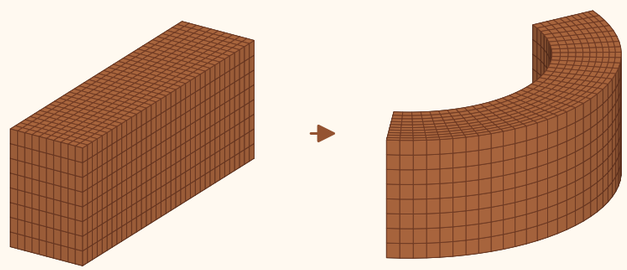
Determine all deformations admitting equilibrium for every isotropic hyperelastic law.

Homogeneous incompressible solids · No body forces

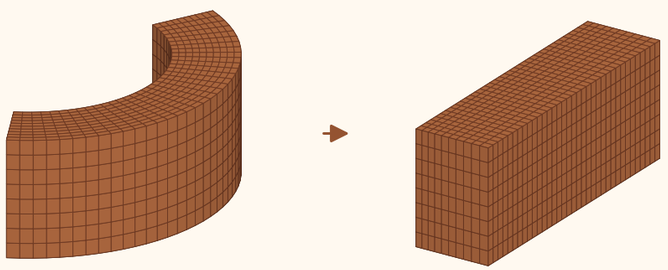
$\mathcal{F}_0$ Homogeneous



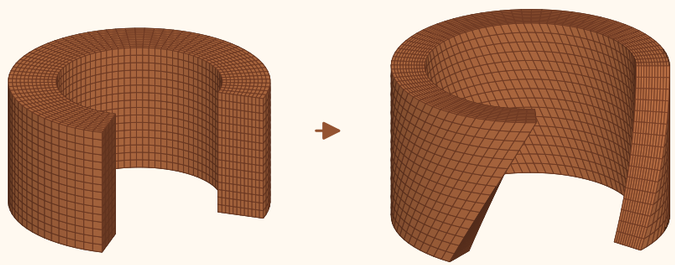
$\mathcal{F}_1$ Block bending



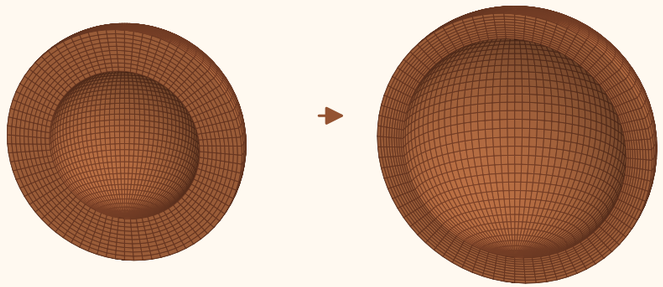
$\mathcal{F}_2$ Sector straightening



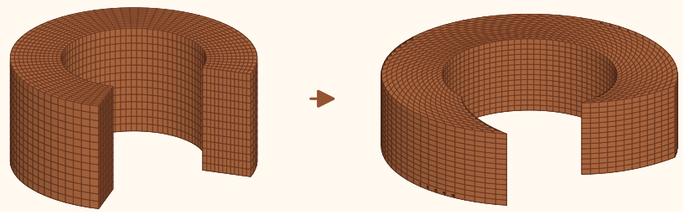
$\mathcal{F}_3$ Inflation and torsion



$\mathcal{F}_4$ Spherical shell



$\mathcal{F}_5$ Logarithmic cylindrical



Complete local classification: the six classical families.

Equilibrium for every isotropic hyperelastic law constrains principal direction fields in three dimensions.
For distinct constant principal stretches, one principal direction is normal to a family of surfaces.